

PRACTICAL NO:19

AIM:MANET Simulation with Directional Antenna

Simulate MANET using directional antennas and analyze improvements in spatial reuse and interference reduction.

1. NS-2.35 was used to compare omni and directional antennas in a 12-node MANET (AODV, 802.11, 4 CBR flows).
2. Directional antenna was approximated using higher gain and reduced power, since NS-2 has no native support.
3. Results: PDR ___% vs ___%, throughput ___ vs ___ kbps, collisions ___ vs ___ (directional vs omni).
4. Directional antennas improve **spatial reuse**, as nodes in different directions can transmit simultaneously.
5. They reduce **interference**, giving fewer collisions and fewer hidden/exposed terminal problems.
6. Trade-off: extra complexity from beam steering and neighbor direction tracking.

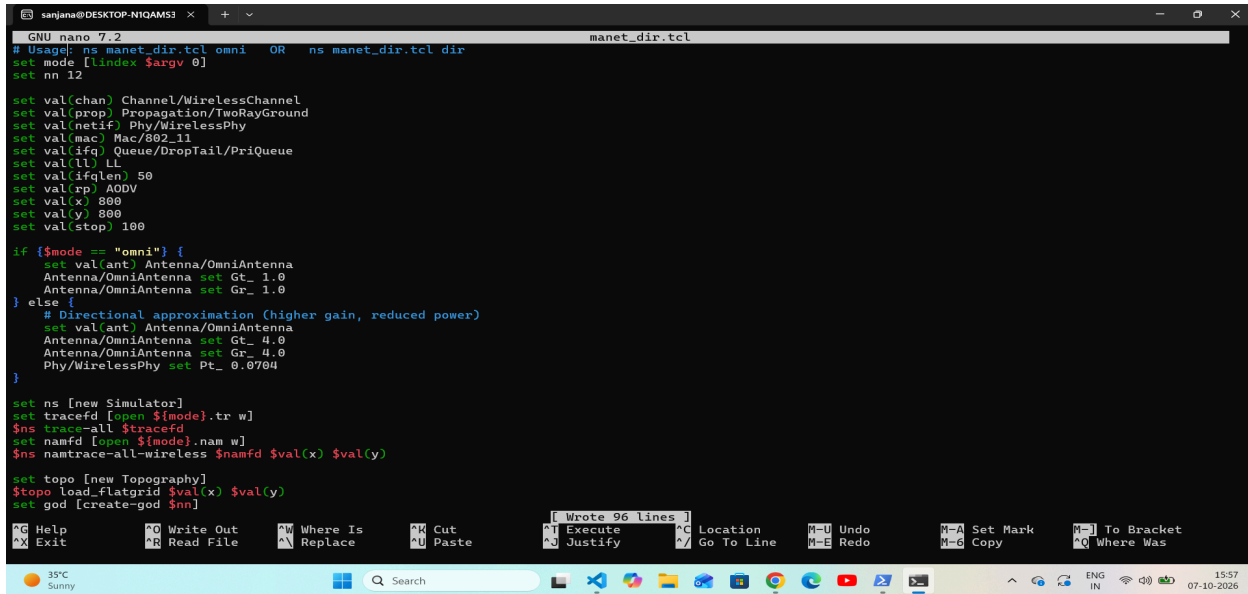
```
sanjana@DESKTOP-N1QAMS3:~$ sudo add-apt-repository universe
Adding component(s) 'universe' to all repositories.
Press [ENTER] to continue or Ctrl-c to cancel.
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
```

```
sanjana@DESKTOP-N1QAMS3:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
71 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
sanjana@DESKTOP-N1QAMS3:~$ sudo apt install ns2 nam gawk
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ns2 is already the newest version (2.35+dfsg-5build2).
nam is already the newest version (1.15-6).
gawk is already the newest version (1:5.2.1-2ubuntu0.1).
gawk set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 71 not upgraded.
```

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```
sanjana@DESKTOP-N1QAMS3:~$ nano manet_dir.tcl
sanjana@DESKTOP-N1QAMS3:~$ ns manet_dir.tcl omni
num_nodes is set 12
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
done
```



```
GNU nano 7.2 manet_dir.tcl
# Usage: ns manet_dir.tcl omni OR ns manet_dir.tcl dir
set mode [lindex $argv 0]
set nn 12

set val(chan) Channel/WirelessChannel
set val(prop) Propagation/TwoRayGround
set val(netif) Phy/WirelessPhy
set val(mac) Mac/802.11
set val(ifq) Queue/DropTail/PriQueue
set val(ll) LL
set val(ifqlen) 50
set val(rp) AODV
set val(x) 800
set val(y) 800
set val(stop) 100

if {$mode == "omni"} {
    set val(ant) Antenna/OmniAntenna
    Antenna/OmniAntenna set Gt_ 1.0
    Antenna/OmniAntenna set Gr_ 1.0
} else {
    # Directional approximation (higher gain, reduced power)
    set val(ant) Antenna/OmniAntenna
    Antenna/OmniAntenna set Gt_ 4.0
    Antenna/OmniAntenna set Gr_ 4.0
    Phy/WirelessPhy set Pt_ 0.0704
}

set ns [new Simulator]
set tracefd [open $mode].tr w
$ns trace-all $tracefd
set namfd [open $(mode).nam w]
$ns namtrace-all-wireless $namfd $val(x) $val(y)

set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)
set god [create-god $nn]
```

```
sanjana@DESKTOP-N1QAMS3:~$ ns manet_dir.tcl dir
num_nodes is set 12
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 388.8
SORTING LISTS ...DONE!
done
```

```
sanjana@DESKTOP-N1QAMS3:~$ ls -lh *.tr
-rw-r--r-- 1 sanjana sanjana 5.9M Oct  7 10:28 dir.tr
-rw-r--r-- 1 sanjana sanjana 4.9M Oct  7 10:28 omni.tr
```

set ns [new Simulator]

Trace files

set tracefd [open manet.tr w]

\$ns trace-all \$tracefd

set namfd [open manet.nam w]

\$ns namtrace-all-wireless \$namfd 800 800

Topography

set topo [new Topography]

\$topo load_flatgrid 800 800

Create GOD

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```
create-god 5

# Wireless channel
set chan_1_ [new Channel/WirelessChannel]

# Node configuration
$ns node-config \
  -adhocRouting AODV \
  -llType LL \
  -macType Mac/802_11 \
  -ifqType Queue/DropTail/PriQueue \
  -ifqLen 50 \
  -antType Antenna/OmniAntenna \
  -propType Propagation/TwoRayGround \
  -phyType Phy/WirelessPhy \
  -channel $chan_1_ \
  -topoInstance $topo \
  -agentTrace ON \
  -routerTrace ON \
  -macTrace ON \
  -movementTrace ON

# Create nodes
for {set i 0} {$i < 5} {incr i} {
  set node_(
i)[ns node]
}

# Initial positions
$node_(0) set X_ 100
$node_(0) set Y_ 100

$node_(1) set X_ 250
$node_(1) set Y_ 200

$node_(2) set X_ 400
$node_(2) set Y_ 300

$node_(3) set X_ 550
$node_(3) set Y_ 400

$node_(4) set X_ 700
$node_(4) set Y_ 500

for {set i 0} {$i < 5} {incr i} {
  node_ i) set Z_ 0
  $ns initial_node_pos node_ i) 20
}

# UDP Agent
set udp0 [new Agent/UDP]
$ns attach-agent $node_(0) $udp0

# Null Agent
set null0 [new Agent/Null]
$ns attach-agent $node_(4) $null0

# Connect
$ns connect $udp0 $null0

# CBR Traffic
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.1
$cbr0 attach-agent $udp0

# Mobility
nsat1.0"node_(1) setdest 300 300 10"
nsat2.0"node_(2) setdest 450 350 10"
```

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```
nsat3.0"node_(3) setdest 600 450 10"
```

```
# Start traffic
nsat1.0"cbr0 start"
nsat8.0"cbr0 stop"
```

```
# Finish procedure
proc finish {} {
    global ns tracefd namfd

    $ns flush-trace

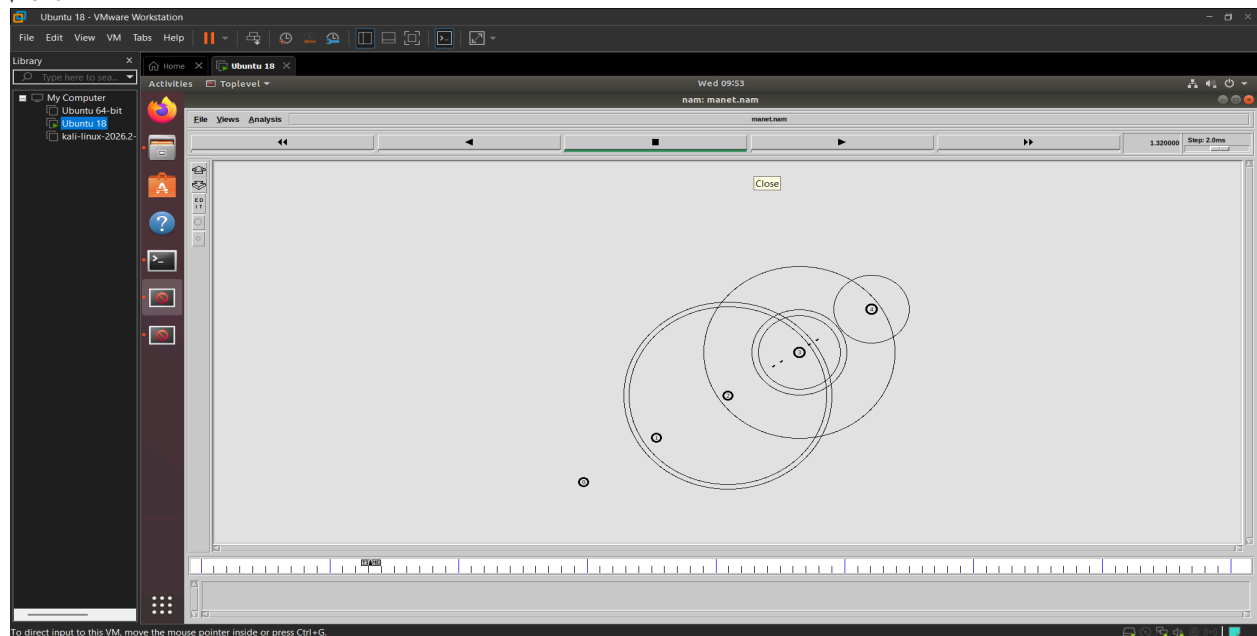
    close $tracefd
    close $namfd

    exec nam manet.nam &
    exit 0
}
```

```
$ns at 10.0 "finish"
```

```
puts "Simulation Started..."
```

```
$ns run
```



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```
sanjana@DESKTOP-N1QAMS3 x + v
GNU nano 7.2 analyze.awk
$1=="s" && $4=="AGT" && $7=="cbr" { sent++; st[$6]=$2 }
$1=="r" && $4=="AGT" && $7=="cbr" { rcv++; bytes+=8; delay+=$((2-st[$6])) }
END {
    printf "Sent: %d Received: %d\n", sent, rcv
    printf "PDR (%): %.2f\n", (rcv/sent)*100
    printf "Throughput (kbps): %.2f\n", (bytes*8)/(90*1000)
    printf "Avg delay (ms): %.2f\n", (delay/rcv)*1000
}

[ Wrote 8 lines ]
G Help Write Out Where Is Cut Execute Location M-U Undo M-A Set Mark M-] To Bracket
X Exit R Read File R Replace A Paste J Justify / Go To Line M-E Redo M-G Copy M-Q Where Was

35°C Sunny Search 16:00 07-10-2026

sanjana@DESKTOP-N1QAMS3:~$ nano analyze.awk
sanjana@DESKTOP-N1QAMS3:~$ echo "--- OMNI ---"; awk -f analyze.awk omni.tr
--- OMNI ---
Sent: 7204 Received: 5762
PDR (%): 79.98
Throughput (kbps): 272.48
Avg delay (ms): 9.20
sanjana@DESKTOP-N1QAMS3:~$ echo "--- DIRECTIONAL ---"; awk -f analyze.awk dir.tr
--- DIRECTIONAL ---
Sent: 7204 Received: 6706
PDR (%): 93.09
Throughput (kbps): 317.12
Avg delay (ms): 29.86

sanjana@DESKTOP-N1QAMS3:~$ grep -E "^D .*COL" omni.tr | wc -l
21
sanjana@DESKTOP-N1QAMS3:~$ grep -E "^D .*COL" dir.tr | wc -l
4080
sanjana@DESKTOP-N1QAMS3:~$ |
```